

Level Design Document

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High Level Overview

Level Goals

Main Objectives

The main objectives of the level's design should be focused on creating an enjoyable and engaging experience. The balance of the map should largely not favour one team, with a moderate amount of balancing to implement this. An overview of the level's main objectives can be found below:

- A complete level experience which functions within the Prop Hunt gamemode for Garry's Mod. This experience must function without issues, players must be able to spawn, turn into a prop, and kept within the bounds of the level.
- The level's experience should be enjoyable to play, with an engaging environment tailored to the gamemode's gameplay objectives.
- The level's balance must be within a fair margin of error of equality, allowing each team a decent chance to potentially win. This balance should be focused for teams of 3, with a total player count of 6 as a baseline.
- A thematic level with visually accordant elements to allow camouflage for the Props team

Secondary Objectives

Secondary objectives are the level's lesser important but ideal objectives that the level must be considering developing towards. They encompass factors less important to the overall design of a Prop Hunt level such as smaller design considerations and visual elements.

- A visually distinctive level with memorable set pieces, level layouts, and visuals all of which is consistent with the Half-Life 2 art style.
- An affordant level which considers the goals of the level and assists player behaviour according to those goals.
- A level that is impossible to exploit through any means, which contains its players well within the level's boundaries, and avoids design pitfalls that allow players to access areas that Hunters will struggle to access.
- A recognisable level which players can identify through the level's unique characteristics that the player can reference through. For example, "The level with the [Blank]"

Design Considerations

Team Considerations

Props

Large levels with plenty of Prop_Physics entities with complex dedicated level space for hiding are the best methods to creating level that appeals to their gameplay loops. Levels with open pathways with plenty of cover preventing large sightlines allow the role to effectively move around the level improving engagement with the gameplay. They're antithetical to the design of Hunters, their beneficial design implements contrast with Hunters which makes it important to balance enough of their needs against the Hunters.

PROPS  Loadout N/a	Character Stats Health - 100 HP [1-200 HP based on prop Volume] Speed - 250 Hu/s Character Information The Props are non-combatant role who's goals are to survive the round without being eliminated. Their ability to turn into Prop_Physic entities allows them to camouflage into their surroundings. They're slightly faster than Hunters and their health is dependant on the volume of the prop they are. 1 Hit point for small props, 200 for massive props. To cater the level for the role, have the level contain Prop_Physics elements. Have enough of them, so that it's easy to blend in, and variety to make it interesting to transform. Make the level have pathways that connect allowing them to loop around the level into different spots to evade capture.
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Hunters

Enclosed levels with a small size, low prop density with simple layouts and open sightlines is the approach to developing the level for Hunters. The timer is their biggest factor for their design, they must be capable of fairly finding Props within the allotted five minutes. The level should avoid design that allows Props to hide away from players without being dealt with. The weapons available for the Hunter role serve different purposes, and their goal is to fill small niches for the role. Crowbars cost no ammo to use, Revolvers are good for accurate, long-range attacks, Shotgun is good for covering space and dealing damage, and the SMG is excellent at dealing with small, moving targets with the AOE and high bullet spread.

HUNTERS



Loadout



Character Stats

Health - 100 HP [5 HP for damaging non-player props]
Speed - 230 Hu/s

Character Information

Hunters are the antithesis to the Prop role. Their goals are to eliminate the Props by using their arsenal of weapons to do so. They're slower than props, and hitting non-player props with damage causes them to take 5 hit points. Killing a Prop restores the hunter who eliminated the Prop player to full health. The weapons a Hunter will have access to is the melee Crowbar, long-range Revolver, SMG/Grenade Launcher hybrid, and Shotgun.

To cater the level to Hunters it's important to consider the timer. The limitation of time means Hunters need a level with a scale and Prop density to match. Additionally, maintaining a level which doesn't allow smaller props to escape to a position where they cannot be dealt with cannot be allowed.

Prop Scale

Prop scale will affect the design and development of the level. The scale of the prop affects the Prop player's collision hull, this will result an increased or decreased in size for the player's collision hull. When it comes to the level's design, larger props will struggle moving around levels that are not built to accommodate their larger frame. Smaller props will be capable of escaping bounds and hiding in smaller spaces, which the level accommodates through blocking player. Designs consider this necessary convention during development to prevent exploitation with the level.

Additionally, player prop scale negatively affects the Prop player's jumping height, players will receive a reduced jump height from all props. As such, the design is to be cautious of trapping players in places they cannot escape from such as pits, and to add ladders to enable vertical mobility for Props.



**Level must accomodate
for all sizes**

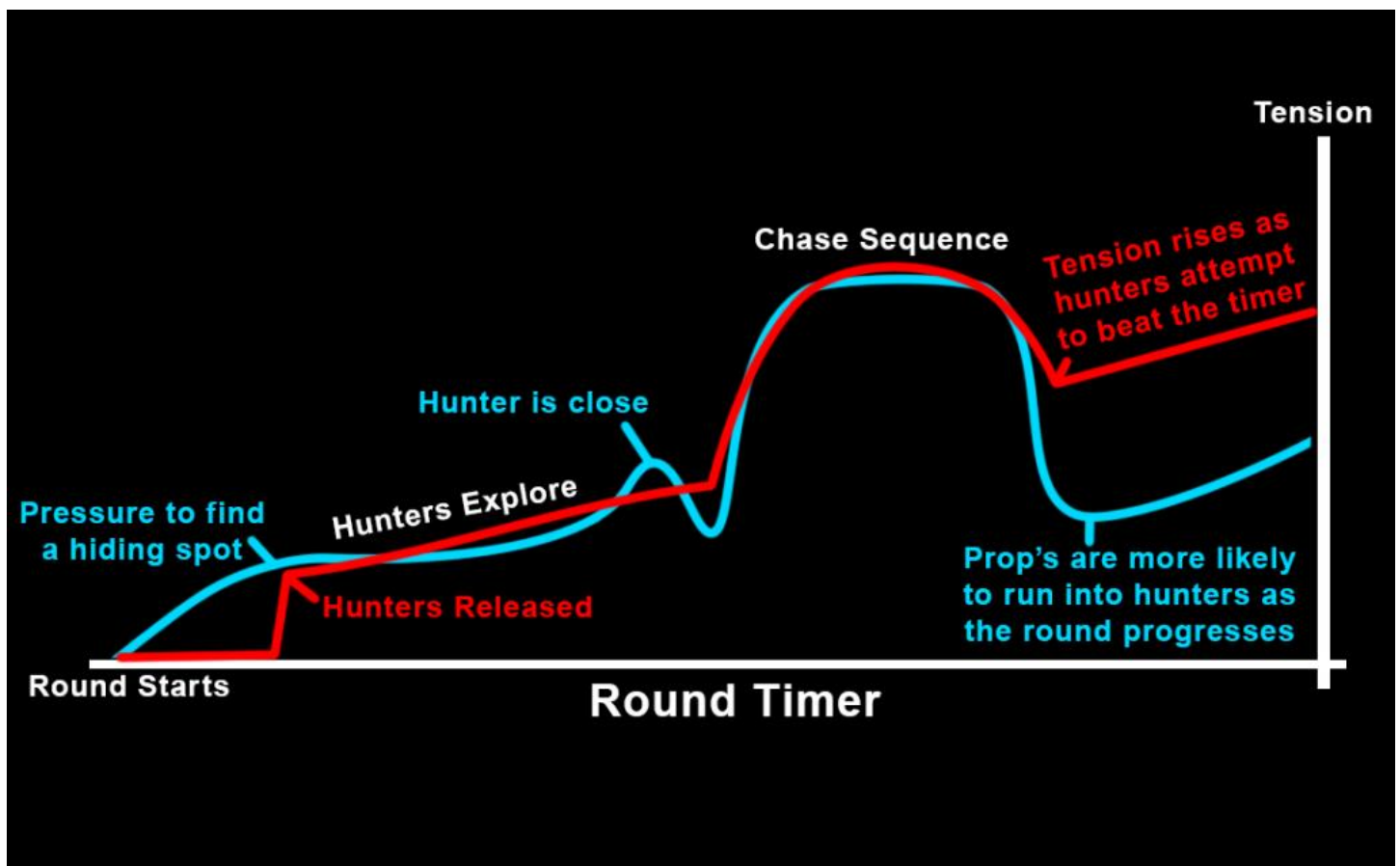
Movement Conventions

The limits of player ability within the level are consistent, neither character role has access to character conventions specific movement options (Outside of small props) that affect the design differently. This makes the level's design predictable regarding the conventions of what players will be capable of performing during gameplay. The only factor that separates the two roles is the default speed, which is slightly faster for Props. This design is intended to allow this team to evade capture as it gives them an advantage when running away from Hunters. Design-wise the level's conventions conform to the capabilities of the player's movement speed and jump heights. There is a minor that Props can use their additional movement speed to make slightly larger jumps. Considering this during development will be important to prevent the Props from escaping the level or reaching an area that the Hunters cannot reach.

Team	Props	Hunter	Description
Speed	250 HUpS	230 HUpS	Players default movement speed.
Crouch Speed	50 HUpS [0.2*Speed]	46 HUpS [0.2*Speed]	Player's speed when crouched.
Health	100 HP (1-200Hp based on the volume of the prop)	100 HP	Player's default health.
Step Height	18 HUs	18 HUs	Max height of an elevated surface to climb without jump.
Jump Height	20 HUs	20 HUs	Vertical height gained from jump.
Crouch +Jump	58 HUs	58 HUs	Vertical height gained from crouching before jumping

Tempo

The level's tempo is consistent throughout each map as the intensity of the gameplay is formed from the gameplay and not the level. Tempo intensifies based on the conditions of the timer, whether it's counting down for a release or win, the anticipation of an event causes tension to rise in relation to the event itself. Player interactions can and will happen anywhere throughout the map, it cannot be predicted, but tension will rise based on types of interactions such as an enemy being near a Prop or evading a Hunter.

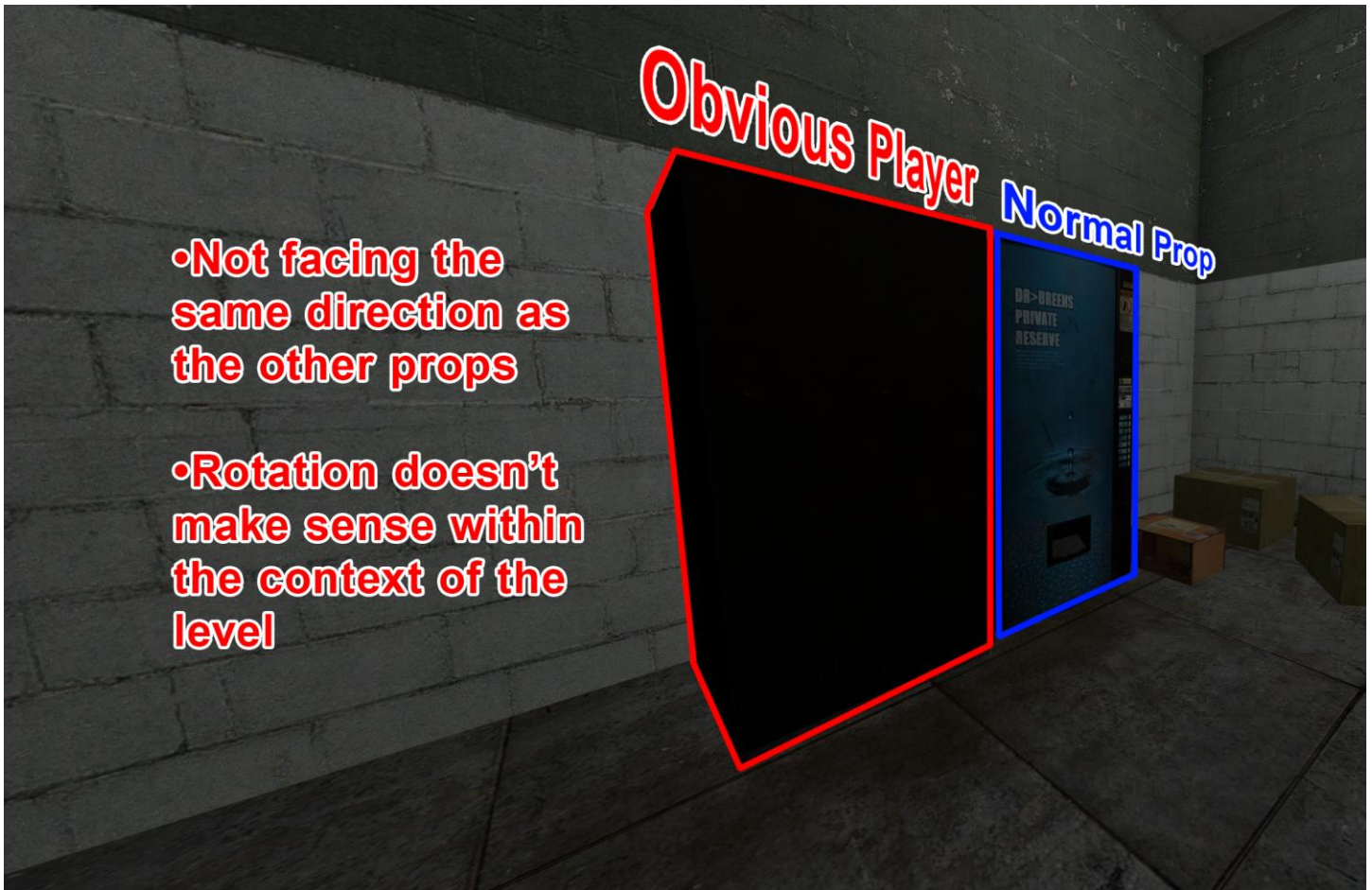


Prop Hunt Conventions

As well as consideration for gameplay and design, further considerations for the level for the elements necessary to allow the functionality within Prop Hunt gamemode. For this, there are two mandatory conventions for the level to work, those being the ability to spawn in, and access to Prop_Physic entities.

Prop_Physic Entities

Garry's Mod's Prop Hunt requires players to have access to Prop_Physic entities to maintain the gamemode's full functionality. Whilst not necessary to get the level running, these actors are required elements of the gameplay loop, and necessary conventions for play. There is the additional factor of the rotation of prop entities within the level and how that factors into the implementation of props. Props players will always assume the 0,0,0 rotation of prop, this is an absolute unavoidable state that the player is unlikely to be aware of. As such, implementation of prop entities should always be at 0,0,0 unless the props are in a group, with no specific direction necessary.



Valid Spawns

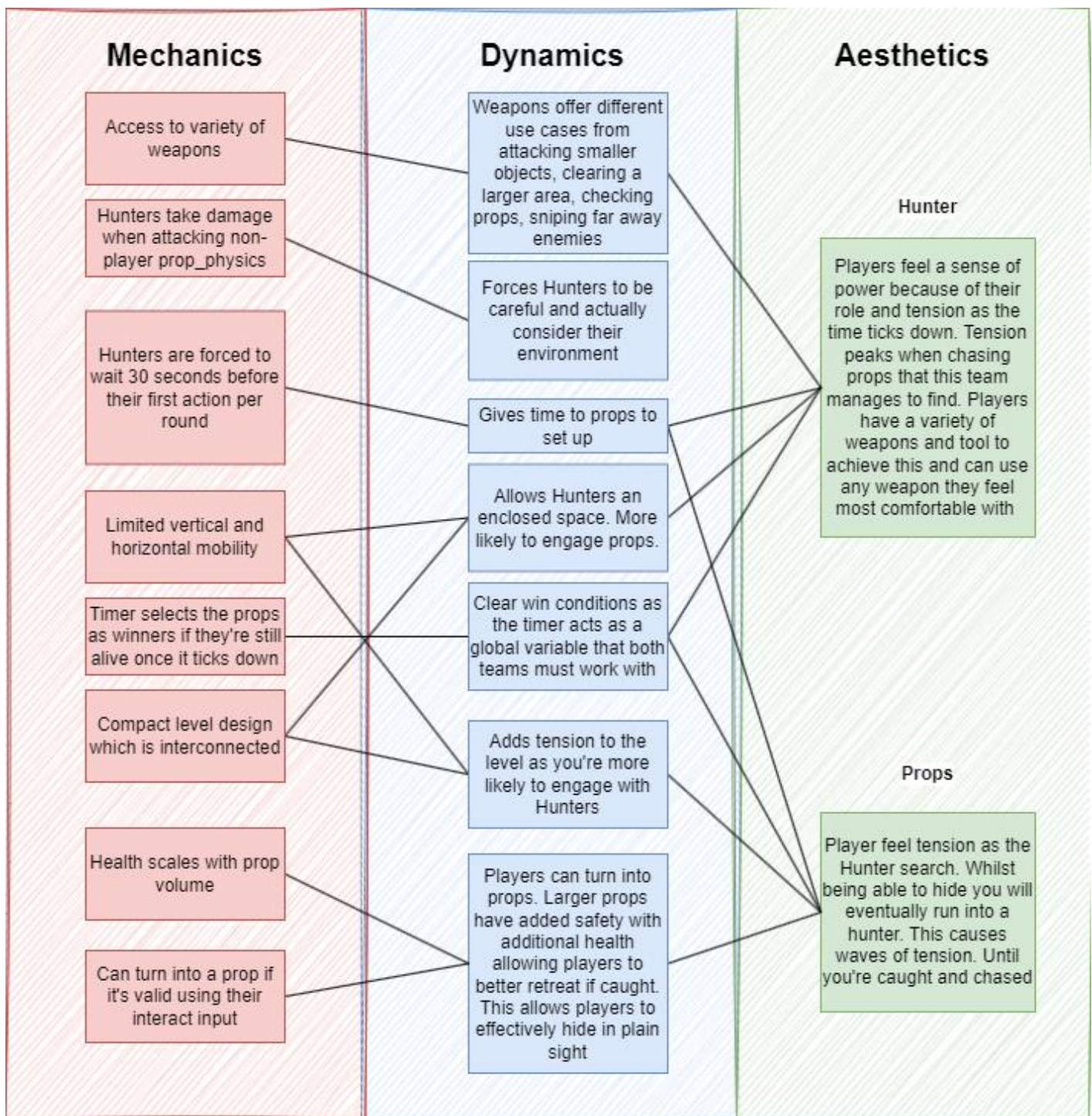
Hammer requires player spawning entities as a way of placing players within a level. This level convention is necessary for the functionality of the level and will not compile without a single valid spawn. When it comes to spawn locations, multiple spawn points must be used to prevent players from spawning within each other. A minimum of 32 spawn points should be used to allow the level to go up to the average server size of a Prop Hunt server.

Gamemode MDA

The MDA for this level identifies several key factors the level must consider when being designed. These factors are necessary for functioning, fair design and cannot be ignore when it comes to current and future design. A full list of considerations can be found below.

- Props need to exist in the level, they need to be valid, and there should be enough of them in one level that you're able to blend in. This is because Prop_Physic entities are needed to be used to transform, and enough need to exist as to be able to blend in.

- Map should be scaled correctly; this ensures that players are regularly interacting. The stage should be scaled up and down based on how quickly the rounds are occurring, player feedback, and the empirical outcomes. This will help to ensure that the gameplay is fun, balanced, and interactive.
- Map should conform to the limited mobility, areas should be clearly navigable for the movement conventions, neither team should have space where the other cannot reach.

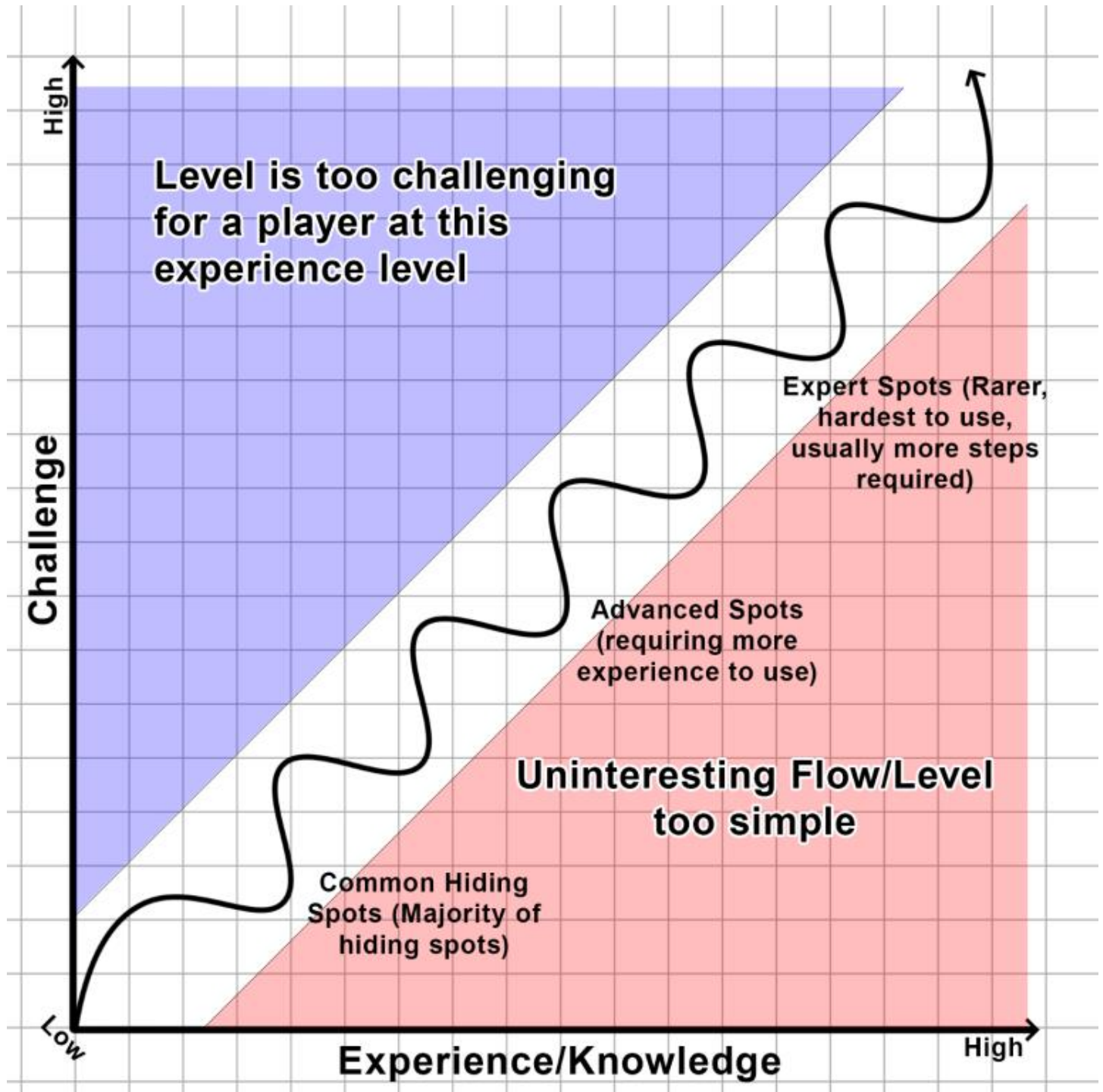


Player Motivation

Player Motivation Overview

Player motivation is an aspect of the level that details how motivation is built for the level, and how the experience should flow for the players. It is necessary for the level to offer increasingly harder hiding

locations as player knowledge and skill improve. The level should avoid making hiding too difficult throughout, with a majority of the gameplay environment accessible by all players. As skill/knowledge increases throughout, the few challenging spots are to become accessible. This motivation chart follows curiosity, and scaffolding.



Explorer

Explorers are the primary social type for the level's design. These individuals enjoy the thrill that comes from discovering new secrets and locations in a level. For a gamemode about finding secret locations and rewarding players to do so, the level's design is built to cater to this curiosity. To do this, designing the level to include advanced, and expert level hiding spots is the way to do so.

Socialiser

Socialisers are a secondary group of people the level is designed to cater towards. These individuals value to the connection that games can bring, and for a gamemode about player interaction smaller design considerations are made to support their experience. Building a level that results in more interaction is the way to design for these players. This can be done through additional pathways, chokepoints, and good level balance.

Balancing Approach

Balancing for the level will be done through two main methods, scale and prop density. The gamemode's timer is an absolute force that all levels consider as a factor for design as its implementation limits the time Hunters can spend searching. As a result, levels need to be designed to allow Hunters a reasonable amount of time to explore and locate Props hiding within. The level's approach is to follow similar examples, specifically for the scale, then balance the level of props based on feedback. This order allows the level's structure and prop placement to be the fully designed in accordance with the guides provided. Prop density, an aspect that doesn't need to design will be adjusted to balance the two teams accordingly.

Map	Size Estimate	Size Estimate Squared
ph_infernaria	3072x2560	2804 HUs ²
ph_office_fsg_v2_d	6144x5120	5608 HUs ²
ph_cornfield_d	3072x4096	3547 HUs ²
Average	3986x3986	3986 HUs ²

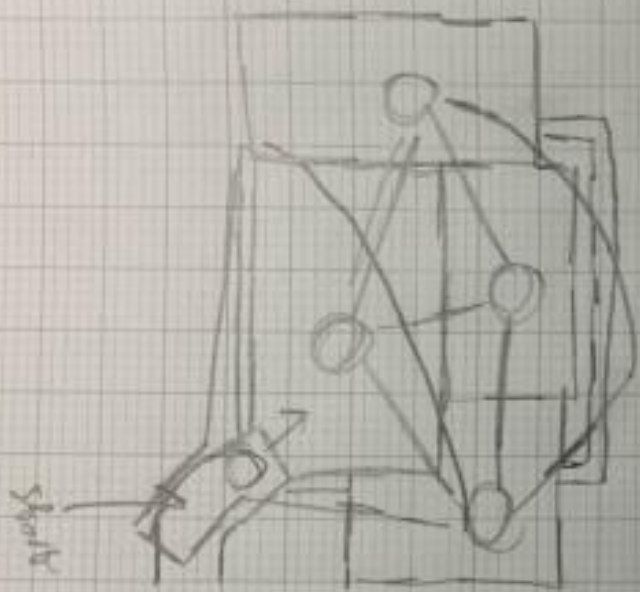
Level Design Overview

Map Structure

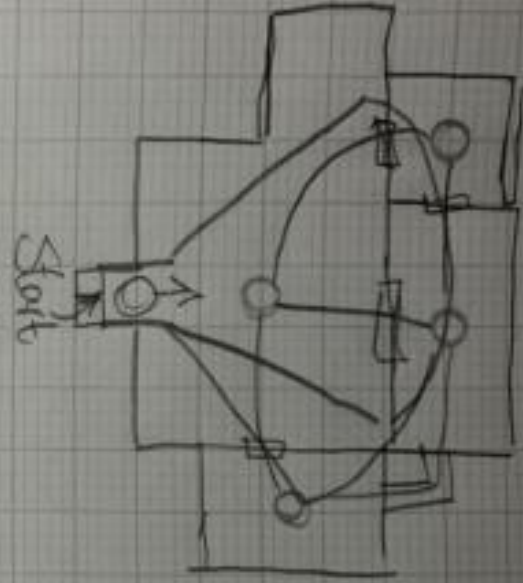
Level Structure

The approach to the level's initial designs were to consider the base level structure with a simple layout concept. This simple concept is then refined into a full-initial design which were analysed, refined and made into the final iterations of the level. Carefully considering the layout is important as the level's design is for an arena-type gamemode.

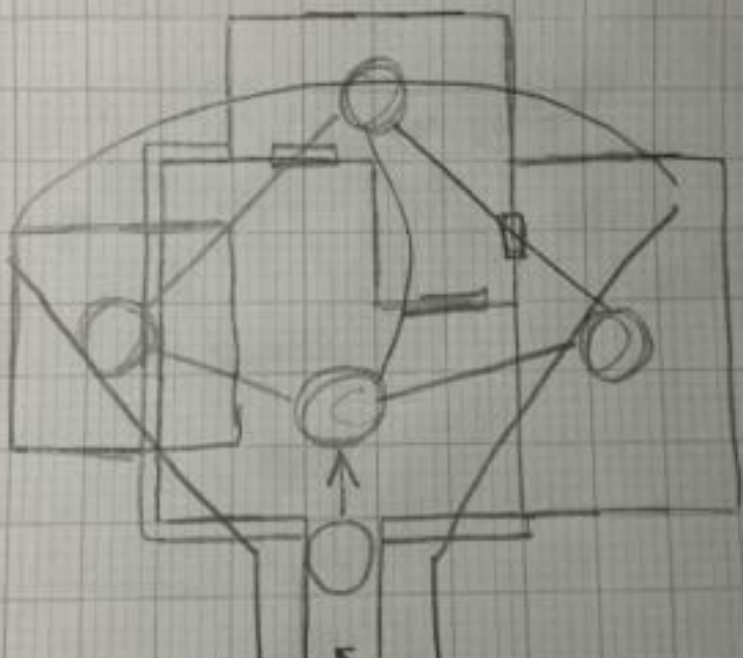
Concept 2



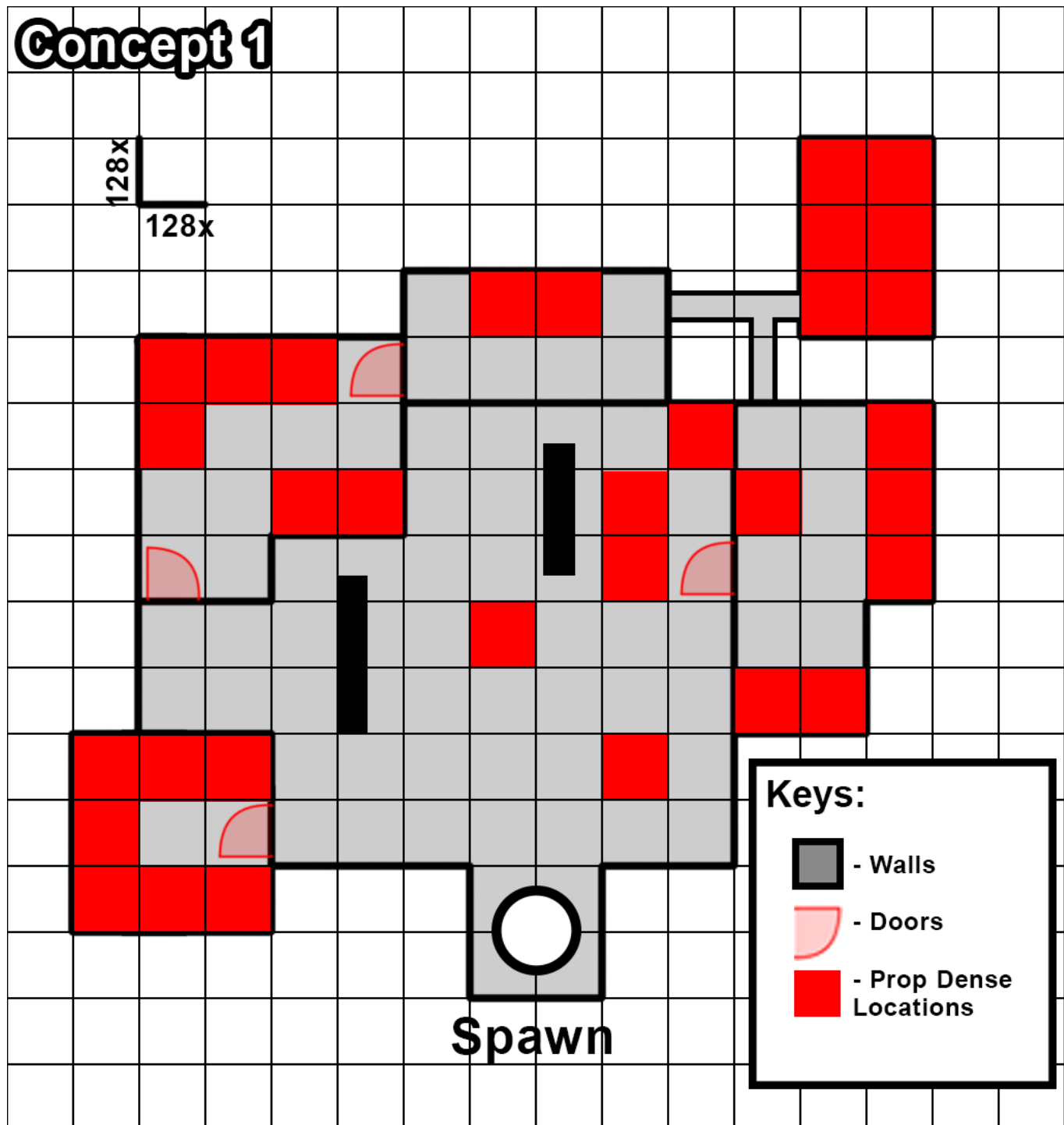
Concept 1



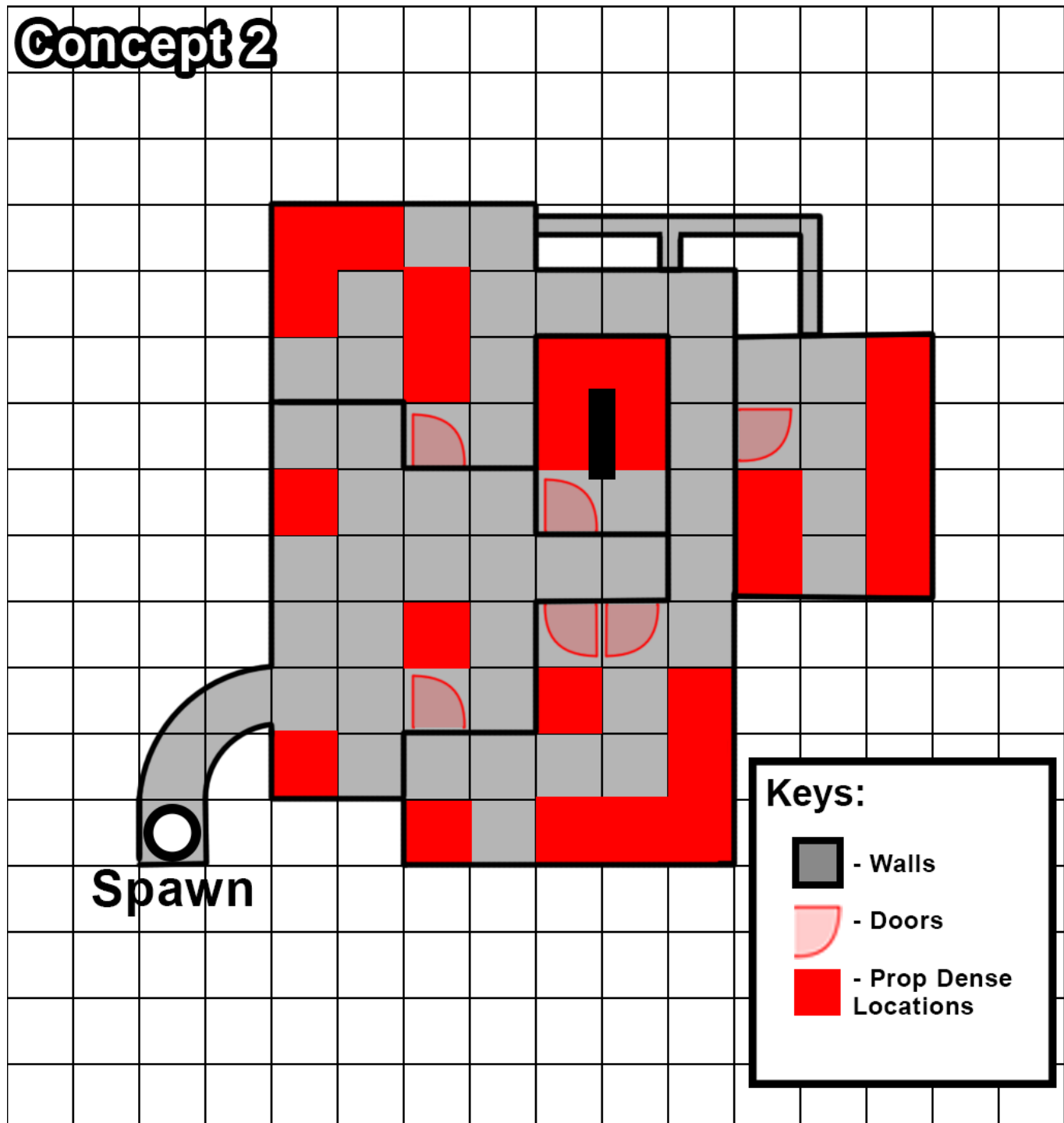
Concept 3



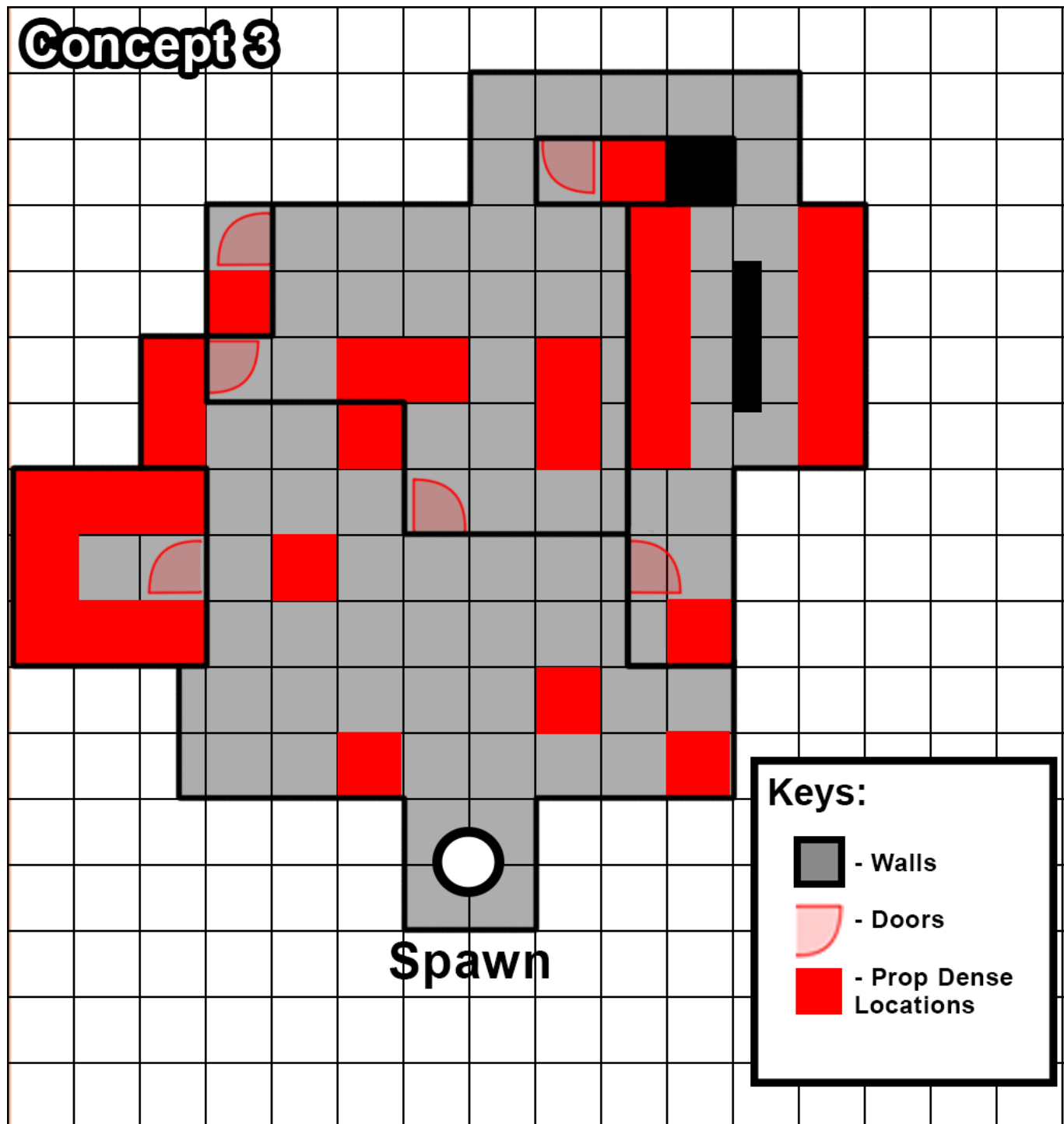
The first concept is unique in its design, due to the spread-out rooms within the level, as well as the differing rooms. Although, the overall scale and openness of the level were poor for the gameplay loop. As a result, this concept's design didn't hit the qualities that the required out of a Prop Hunt level. The unique pathways and rooms were a consideration in further iterations such as the vent found in the final version. The 128x represents the HUs scale to grid size and is the baseline scale for all the initial designs.



The second initial level concept is focused on improving on that, the level is comparable more compact, and inside, so the level doesn't benefit Hunters. The layout is much more connected with levels usually connected by two or more pathways. The smaller scale does benefit Hunters, which could cause issues, it could have more scale to it. Additionally, the flow is lower due to the long corridors and the map doesn't work well with the tutorialised funnel which could make the level worse for new players. As such, the level wasn't used for the blockout.



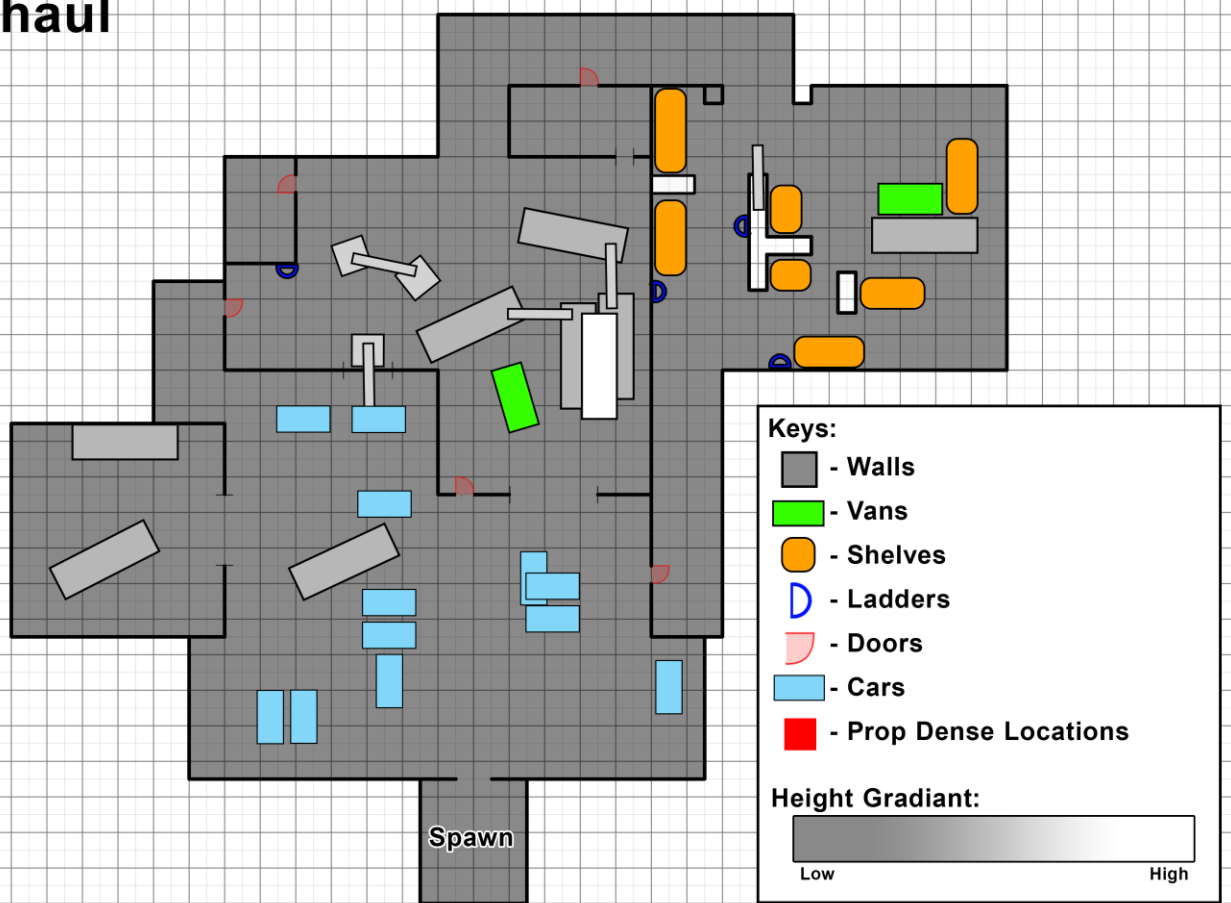
The third and final initial design balances the elements which the previous examples did not effectively use. The scale and openness of the level struck a good balance throughout to best work alongside the intended balance of the level. The level implements tutorial techniques within the structure. These factors make the concept the best at working with and as such the level was selected for the blockout.



Improved Design

The improved design follows the basic initial iteration chosen and refines it for with updated inclusions of further considerations. Improving on the aspects in the initial design were the inclusion of necessary conventions (Doors, Ladders, etc), further refinement of the layout, and changes made throughout development. Changes such as making the outside less open, adding additional pathways in the form of vents are the changes made to this version-based elements of design found within other initial design. Everywhere outside of the boundaries should be out-of-bounds, inaccessible for the player(s) without the use of exploits.

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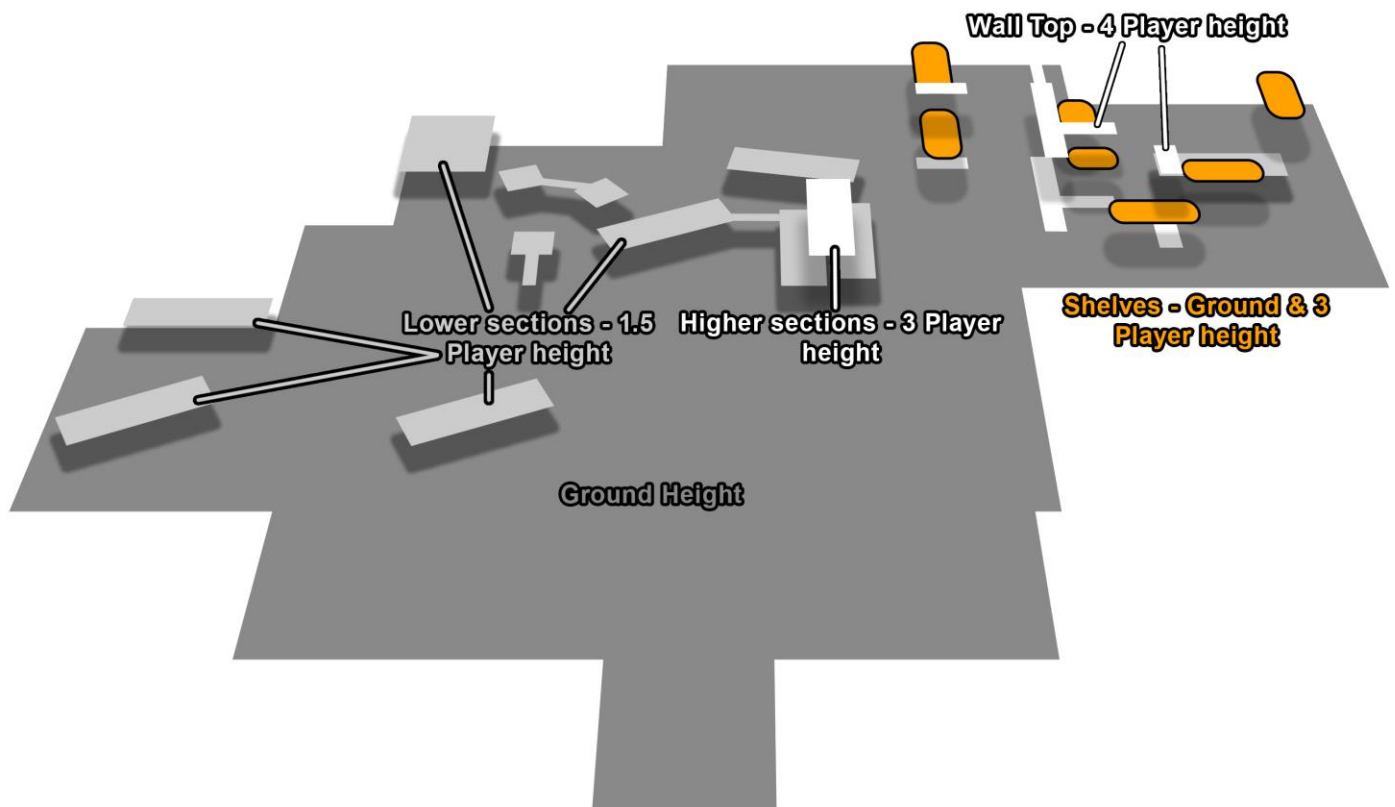


Layout

Elevation

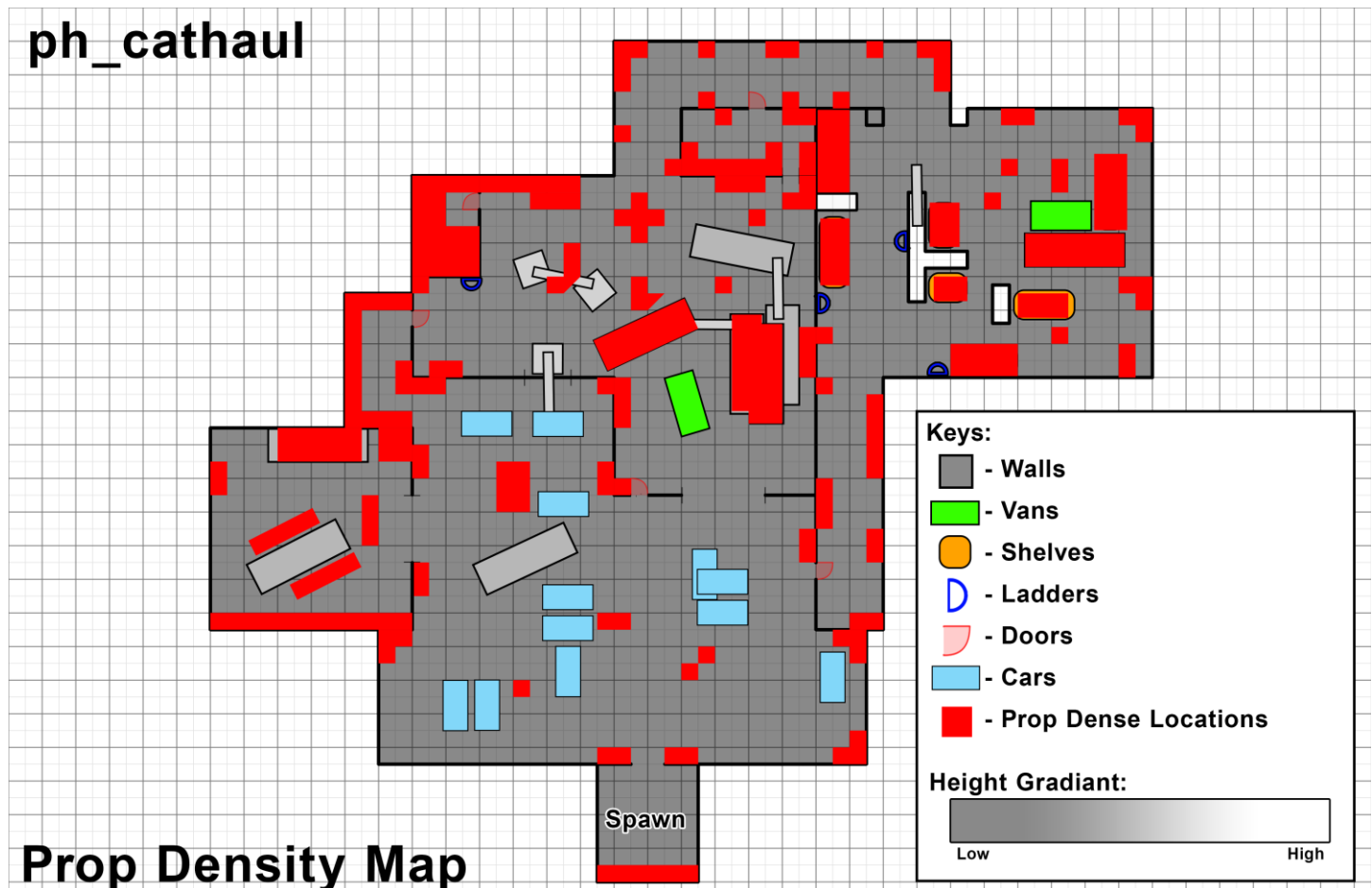
Elevation has been an element of consideration throughout the development of the level. The details on the original improved design layout further; this map is instead a better rough estimate of the height of each portion of the map's macro elevation (Elevation that matters to the design, not elements such as single steps or minor terrain differences). Increase elevation does not bring additional balancing

concerns, not any more than adding new space to the level. Designers and developers need to consider how players may interact and use the new space. Does the added elevation add new exploits, is it properly accessible, if so, why and how? Elevation in its current state is designed within those considerations, Ladder make the area accessible, the space does not bring along new exploits (enclosed space), and the design is tailored towards novice-expert knowledge levels.



Prop Layout

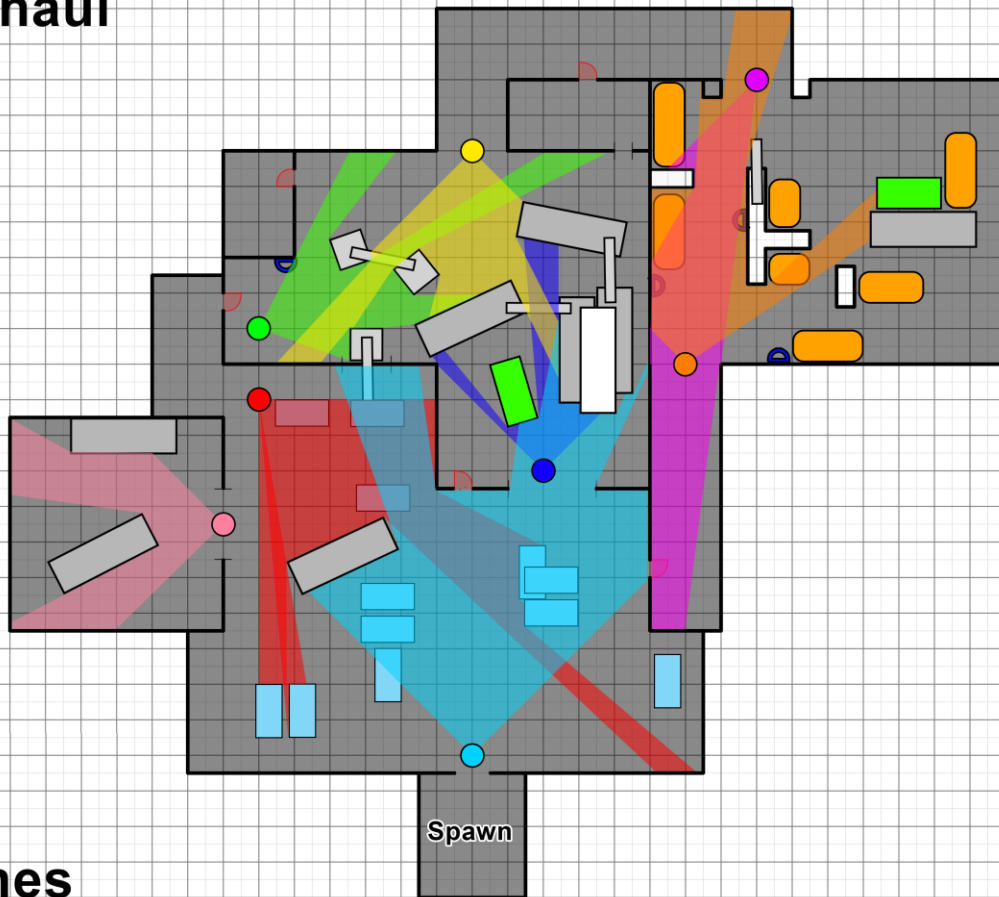
The level contains a variety of props throughout the level as a necessary convention for the gameplay loop and Prop role. As an essential, plannable element within the level a map of the layout of the props within the level. This map acts as a guide for where props should be placed throughout the level, with at least one Prop per tile up to the balance cap of the map's prop balance limit. By having this map, developers and environment artists can implement identify where Props are going to be, where they need to be placed. Props are not placed on pathways, allowing the player to move around props within colliding throughout each section of the level.



Sightlines

Sightlines are a minor consideration for the layout of the level as Hunters cannot have access to large open areas. In a gamemode about finding enemies, a large open sightline makes that typically easier. Even still, it's only a minor consideration for balance to encourage a level which is designed to allow for Props to move from locations. In general, the layout being enclosed throughout, with a focus on non-open areas means the sightlines are balanced. This map showcases how the map has been tested to prevent sightlines from becoming too powerful and allowing Hunters to dominate in this regard. The overall layout, sharp corners and heavy usage of walls is the reasoning behind sightline balance.

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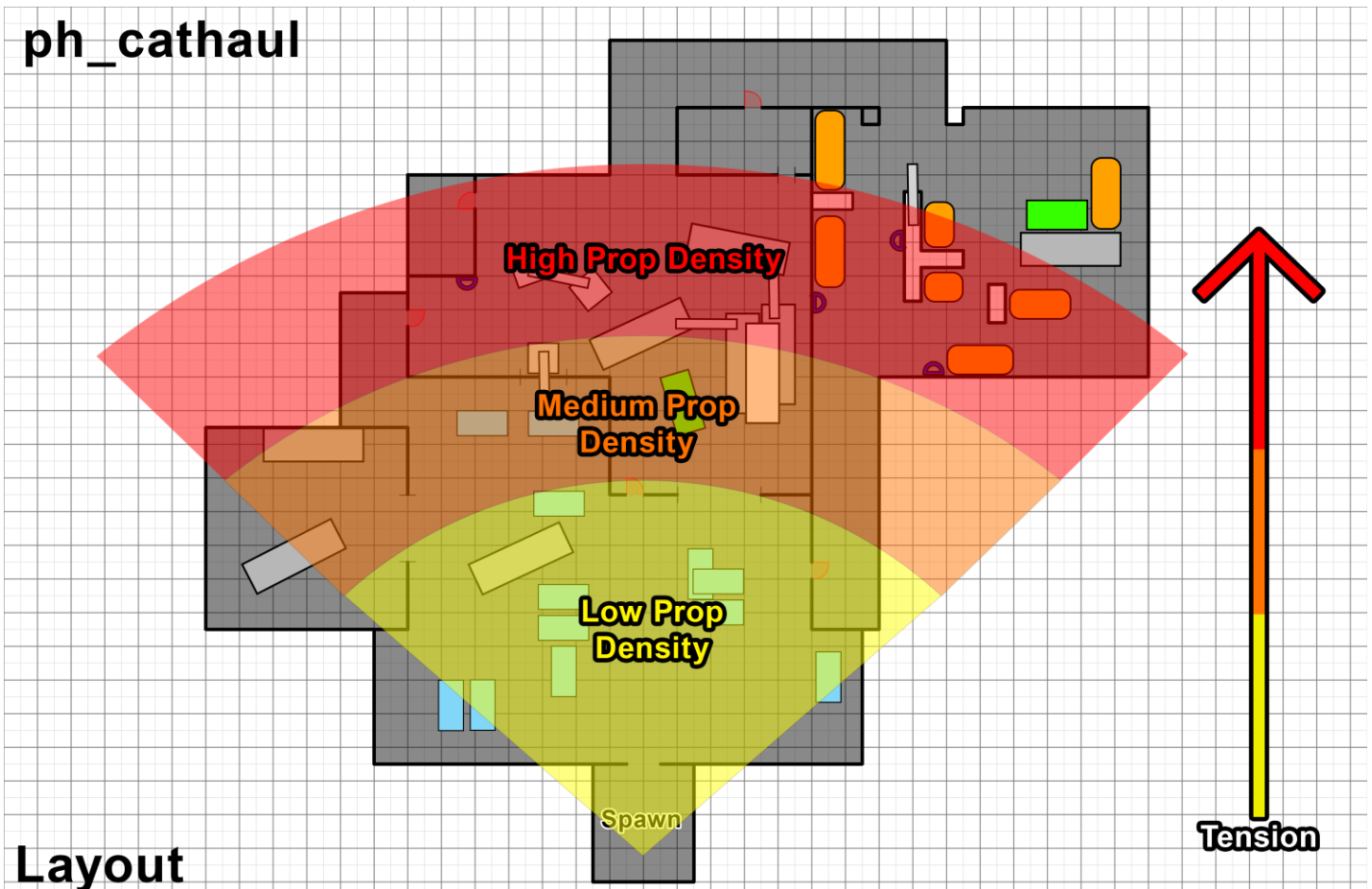


Sightlines

Level Tempo

The level's tempo is unique in-terms of implementation, but the tempo still exists within the level in a different context. The implemented and ideal tempo for this level is one which has an increasing quantity of props the further into a level the player is. Depending on the intensity of the area, the average density of prop entities is to be higher or lower, all within the density range. A map detailing the density can be found below.

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Blockout

The blockout phase focused on building in the important aspects of the level's intended design. As such, visual elements have been implemented, alongside with the layout to enable props the ability to blend in with their environment. Overall, this build achieved what was necessary out of the level, A semi-balanced experience that conforms with the conventions of the Prop Hunt gamemode that is fun. This version was subject to changes, which helped to refine the issues and improve the experience for players, these are the early renders of the level.

Over-Head View



Beauty Shot



Level Final Iteration

Spawn



Changes

- Added additional lighting to the space to make the area easier to navigate and to highlight the sign better
- No major changes made.

Carpark

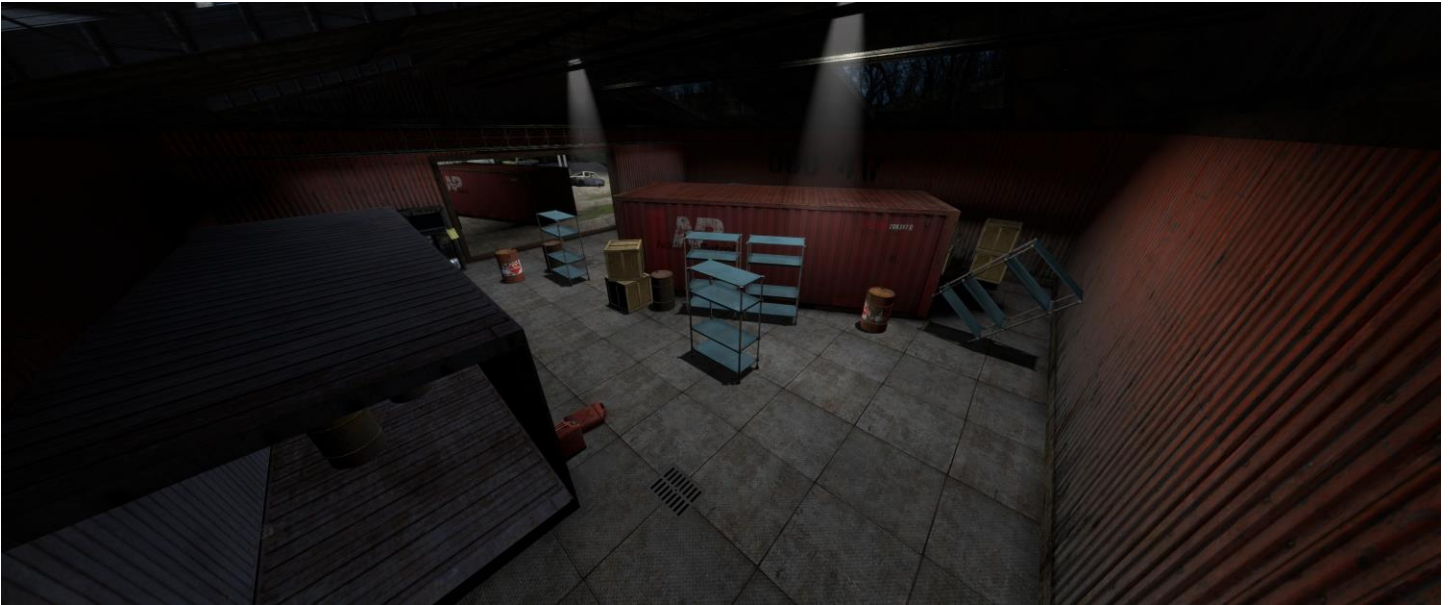


Changes

- Added shipping containers and opening to Main Green Warehouse to improve player pathing and encourage players to move towards the Red Shed.
- Level section visually polished.

- Prop density reduced, with props being reshuffled to act more affordant with the usage of breadcrumbs to assist player pathing from spawn.

Red Shed



- Added affordances outside the shed to help direct players towards this direction.
- Prop Density reduced throughout section slightly to adjust the balance.
- No major changes made

Changes

Main Green Warehouse



Changes

- Added additional pathway with a hole in the wall to encourage players to head towards the red shed and outside areas.
- Prop density reduced in accordance with balancing.
- Van given player clip to prevent exploitation under the element as a small prop, becoming difficult to find for Hunters.

Main Warehouse Office



Changes

- Prop density reduced slightly to improve balance and movement within this room.
- Corpse added to improve narrative through environmental story telling.
- No major changes made.

Trash Pile



Changes

- Prop density of smaller props heavily reduced to make exploration of this smaller area easier and proportional to the size within the level.
- No major changes made

Back Hallway



Changes

- Reduced prop density to match the importance of the area and to further balance the level.

Backroom



Changes

- Floor changed to differentiate space against the different level backdrops.
- Added additional pathway in the form of an annotated vent between the shelves to add the options player have when using this space during gameplay.

Orange Secondary Warehouse



Changes

- Entire room reshaped and expanded horizontally at the cost of verticality to improve gameplay and movement within this space.
- Lighting improved to make the space more visible for players and to reduce the ability of players hiding in corners
- Improved the area's tougher hiding spots within the level to improve overall gameplay flow.
- Van given same blocking volume treatment as the other van to prevent players from abusing it as a small prop.
- Ladder's end points improved to prevent potential issues players may have with them.

- Space made to be more affordant through the changes made to the wall textures to give this space a distinctive orange colouration.
- Props density is rebalanced and spread out over the entirety of the space to improve the balance and to spread out the props within the room.
- Added anti-affordances alongside doors to improve the level's themes without impacting the design.
- Corpse added to improve narrative through further environmental storytelling.

Front Hallway



Changes

- Hallway changed visually to differentiate itself from the rest of the level similarly to the Orange Secondary Warehouse
- Added anti-affordances to the additional artistic implementation of the door within this space to build the theme without affecting the design.

Overview



Changes

- Added NoDraw brushes to sections of the level where faces aren't visible at all to the player to improve level performance.
- Skybox added to support the level's atmosphere, tone, and themes.

Beauty



Points of Interest

This section discusses the different implementations of locations within the level to showcase their identifying features that players can use as a means of quickly describing an unknown location. This is particularly useful for player who are using call outs, which is likely to occur in this team-based gamemode. There are four main locations players are likely going to call out. To differentiate these areas there are specific implementations to do so.

Car Park

The identifying features are in its uniqueness within the level. There is only one car park so players can differentiate the area by this aspect. Players will likely callout this location with descriptions such as “Outside”, “Carpark”, “The area with cars”. This seperates the area from the rest of the main four points of interests.



Red Shed

Being red and away from other structures, as well as the only small structure outside allows this space to be identified and called out on these obvious and apparent descriptors. Colour also works to help differentiate the space from the other locations with red being uniquely tied to this location. Players are likely to callout this location with “Shed”, “Outside hut”, “Small outside building” “Red building”.



Main Green Room

The main green room's biggest factors are its abundance of shipping containers, the colour of the room itself, and the position within the level. Colour is self-explanatory, shipping containers are numerous in the room and is a potential identifier for this space, and the position within the map is central so players may use that to describe the space. Players are likely to describe this section as "Big green room", "The room in the centre of the map", and "The one room with loads of shipping containers".



Secondary Orange Room

Lastly, the orange room in the back is identified similarly to the main green room, as a large, coloured space with specific types of an element within the level. Colour is yet again used to differentiate, shelves are found within the space so players might identify the location as such, the elevation is very high which is a possible way of distinguishing this space from others. Potential callouts include “Orange back room”, “the room with all the shelves”, “the room with lots of elevation”.



Character Experience

Props

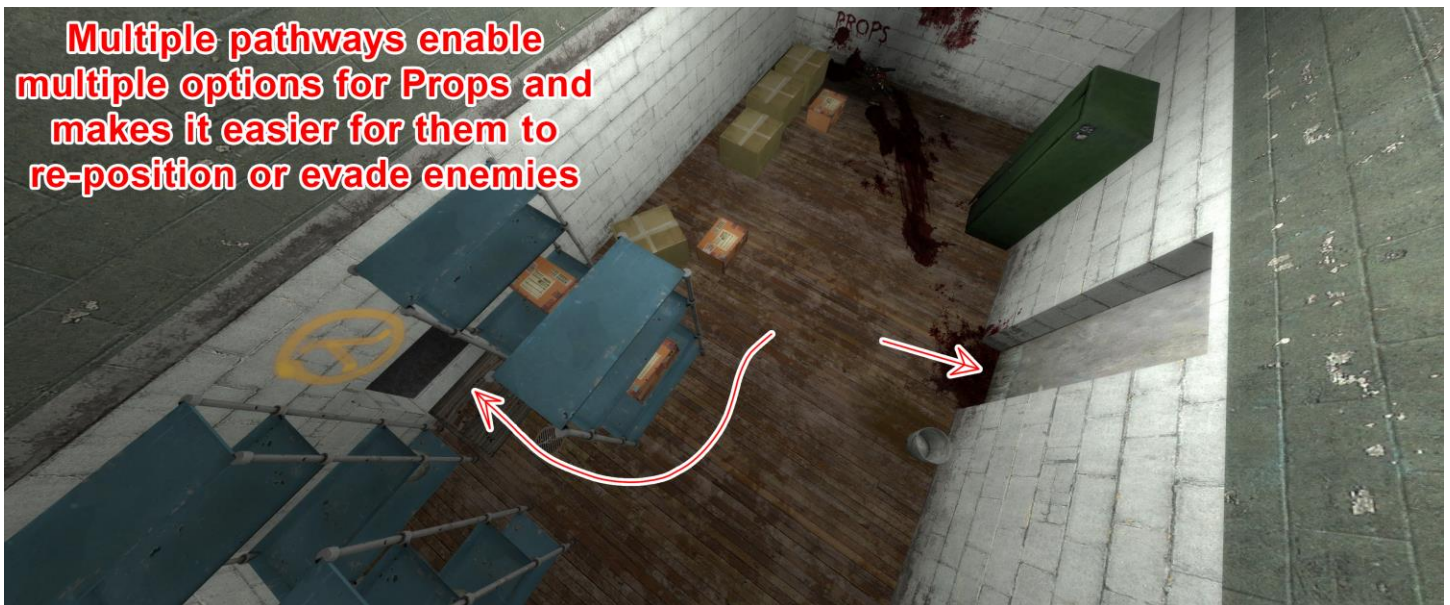
One of the two main character experiences that the level has designed around. The implementations of Props within the level are both done to assist and balance the character role. Within the level, several helpful considerations have been made to achieve their goals based on their typical experience. A Summary of a typical experience for this role goes as follows:

- Spawn in, with 30 seconds worth of set-up. This allows Props to find an initial location to conceal themselves within.
- Props will then lie in wait for Hunters, perhaps they will move around the level in search of a different spot to better evade the enemy.
- It's likely that they will be caught, and a case ensues, the player will run throughout the level in their best attempt lose the chasing Hunters.
- If the player hasn't died yet, and the timer finishes counting down, then the Props win

The expected route of a player throughout the level and the accommodations made are: The player will leave spawn in a direction, heading towards the highly dense sections of the level in search of a hiding place.



Props like to move around and evade Hunters, either on their own will or forced to by chasing enemies. The level supports this with the interconnected level which allows players to move semi-freely throughout the level with multiple available pathways in each area. This helps to also encourage more interaction and thoughtful play boosting the engagement.



Map is littered with similarly thematic props which allows players to camouflage in most environments throughout the level assisting their re-hiding if they've been chased.



Hunters

The Hunter role is the antithesis of the Props, who search instead of hide. Their implementations focus on balancing the role to support their ability to find Props within the preset time. A summary of their gameplay will highlight their goals is necessary to showcase changes to support their implementation:

- Hunters wait for Props to set up, they cannot move and see for the initial 30 seconds.
- Hunters will then head into the level to search for Props, they will navigate throughout the level and check prop entities.
- If a Hunter finds a Prop and gives chase, chokepoints make it easier for Hunters to enclose and catch stray Props throughout the level
- Hunters should reasonable be capable of finding all players near the end of the timer.

Hunters follow a similar expected route, instead their goals are to find players, and they can do that after the set-up time for Props ends.



The level's layout creates chokepoints and slow down sections which allow Hunters to catch out moving props. The generally compact environment further enhances this which balances the strengths of the multiple paths for Props and allows Hunters to benefit from aspects of the level's design



The overall scale and prop density is of a scale that Hunters can find tolerable and fair for them. The outcome of rounds doesn't heavily favour props because the layout has a good balance of the previous two mentioned factors.

Level scale and prop density allow the Hunters to reasonably find player within the time limit



Tutorialisation

Tutorialisation Overview

Tutorialisation throughout the level is limited due to the nature of the gamemode. The level will be replayed several times over, as such the level's tutorial reflects this with limited implementation; tutorials cannot impact player's time, it cannot be a forced section of gameplay. There are two instances of tutorialisation within the level, the sign at the spawn location, and the cones that surround the spawn. These work to establish the concept of Prop Transformation, and the cone act in isolation as the only interactive as well as a component on the sign. Funnelling of the level through the shape that expands outwards from spawn reduces player options which supports player navigation.

Sign acts a signifier of the main gameplay ability for Props

Nearby cones found on sign are used in isolation to encourage players to interact

Level's funnel shape built to make level affordant for players when initially starting

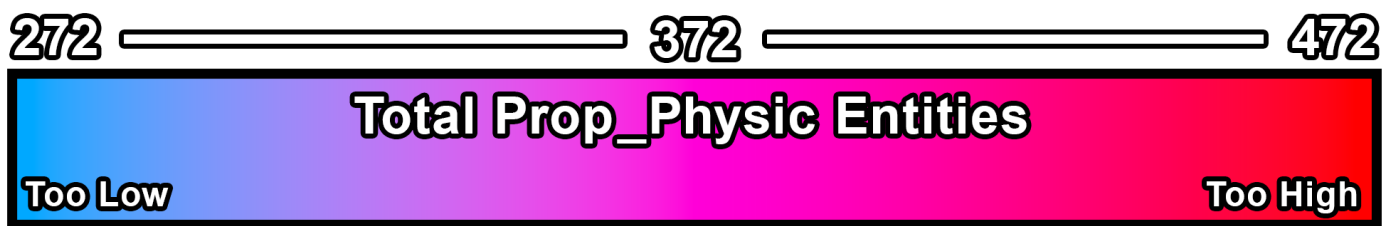


Balance

Prop Density

The density of the level's props is directly proportional to the level's team balance. Higher numbers of Prop_Physic elements within the level's current scale will favour the Props, whereas the opposite applies with a decreasing amount to Hunters. The variable amount difference between iterations can be used to get a rough estimate of the ideal range of Props that should be available within the level's boundaries.

Prop Density Range



Map Scale

The level's scale is another element of balance to follow with when it comes to iterations and the creation of the level. The factor of time limit Hunter's ability to search for Props and creates a hard limited to follow. A mixture of props and scale is used to create the balance as to ensure a level has room for exploration and props to transform into. Larger levels benefits Props, whereas a compact map favours the Hunters. In general, the scale has remained the same as to not add too much difference in balance.

Visual Design

Half-Life 2 art style

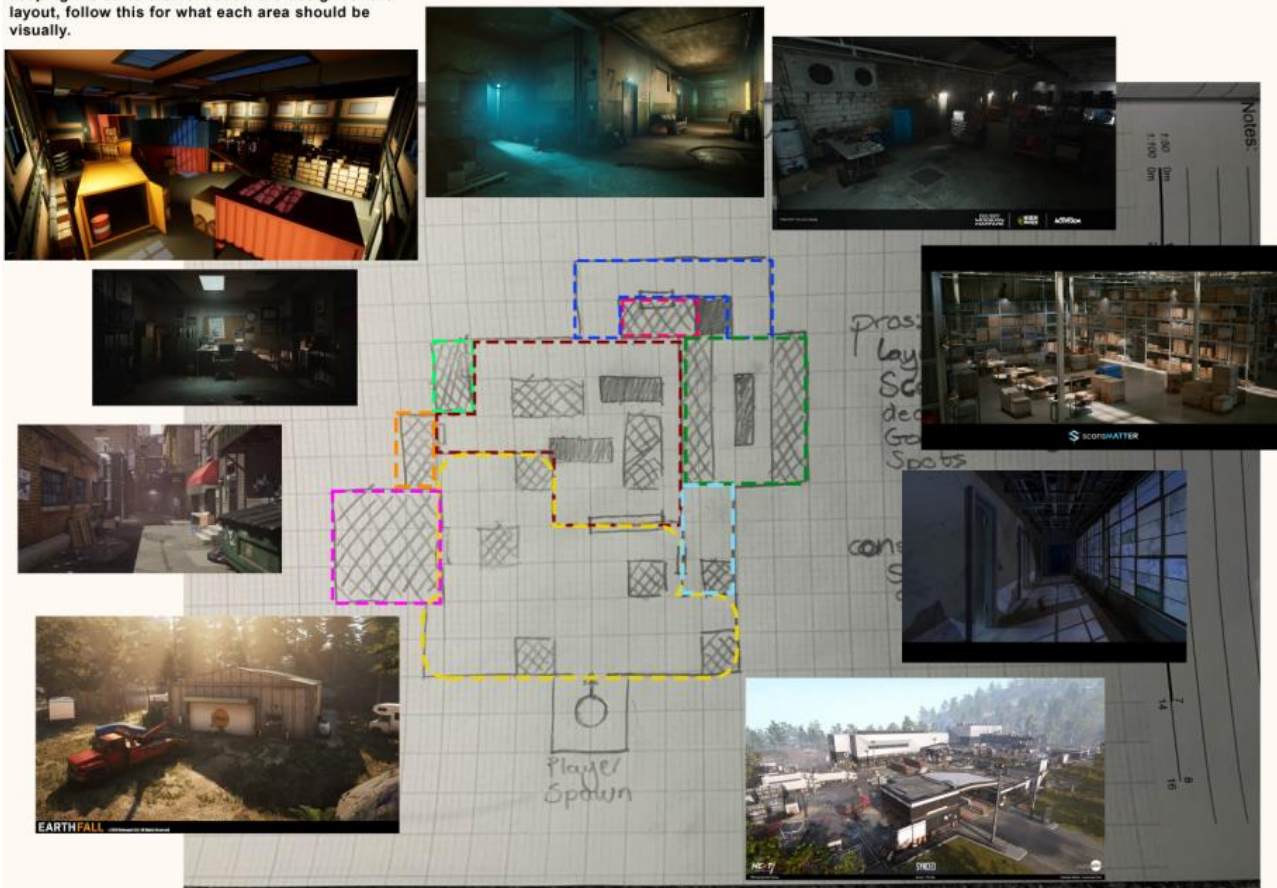
The visual design of Half-Life 2 is a mix of realism and sci-fi themes which amalgamate to create a dystopian, decaying world with humanity in an oppressive police state. The area of the game the level takes the most inspiration are the coastal levels, which are where the majority of textures and assets for the level originate from. There are no specific rules to the art style, the art will be valid if it follows the themes.



Level Visual Consideration

The level visuals have necessary designs that the level follows which form the basis for the considerations for visuals of regions within level. This piece of documentation showcases imagery which is to be followed regarding how to visually develop for each portion of the level. Useful guides for designers and environmental artists to communicate their implementations.

Images shouldn't be taken literally. They're references to what areas should be heading in terms of usage and visuals. Everything should follow the basic Half-Life 2 artstyle when possible and should use assets from that project when developing these areas keeping the same theme. Follow the design for the layout, follow this for what each area should be visually.



Level should have elements of overgrowth
Level should take own liberty when it comes to visuals and follow roughly here

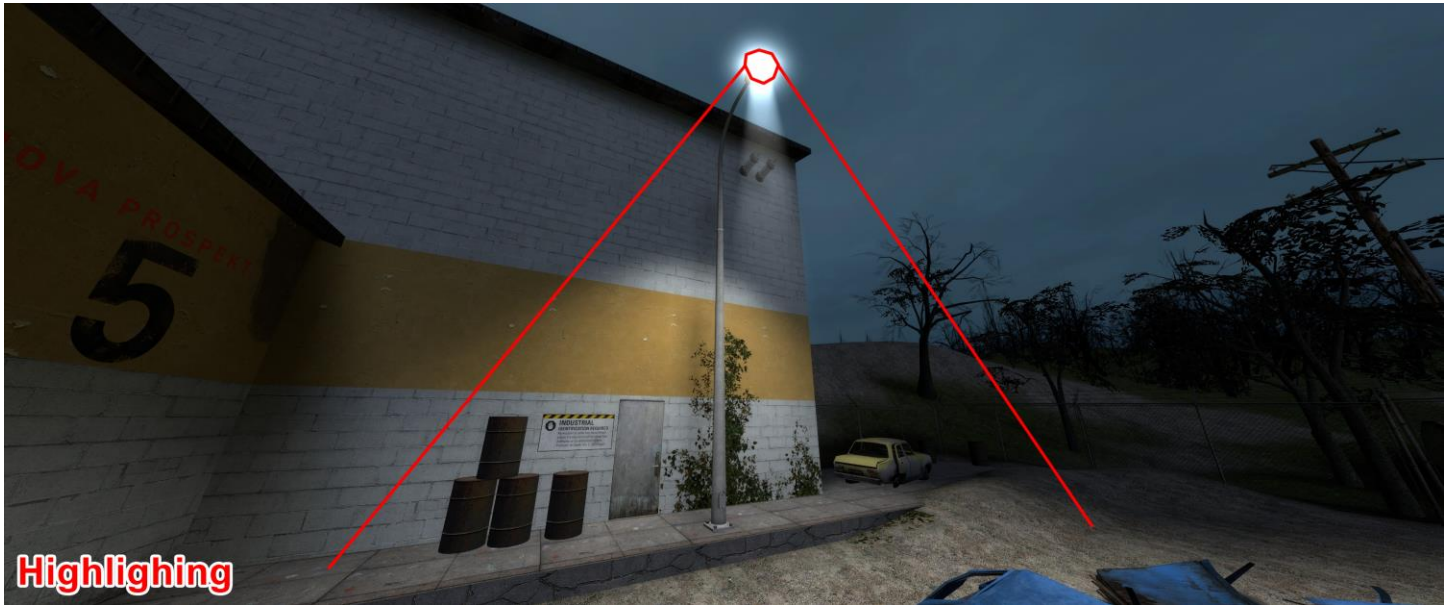
Affordant Design

Affordant Overview

The goals of affordant design within the level is to guide player throughout the layout by highlighting key pathways, interactables, and rooms. This is achieved through subtly, as not discourage players from exploring the level's environments. Implicit encouragement is the best approach for directing player behaviour, due to the casual, visual-focused gameplay.

Lighting Affordance

Lighting affordance is an aspect used within the level due to dark global lighting of the level. Lighting affordance is used in a limited context to not overwhelm player behaviour, as result examples are not massively contrasting. Light in the level is affordant to the design by acting as breadcrumbs or to highlighting key elements, supporting player behaviour towards key areas.



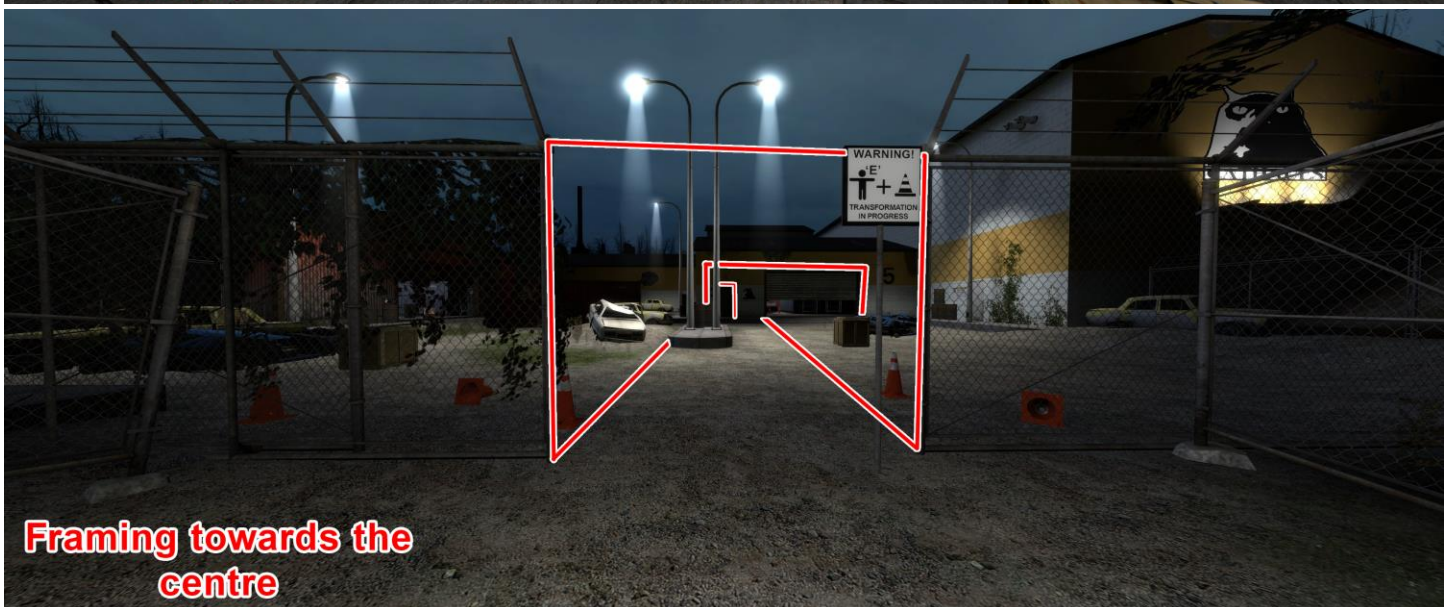
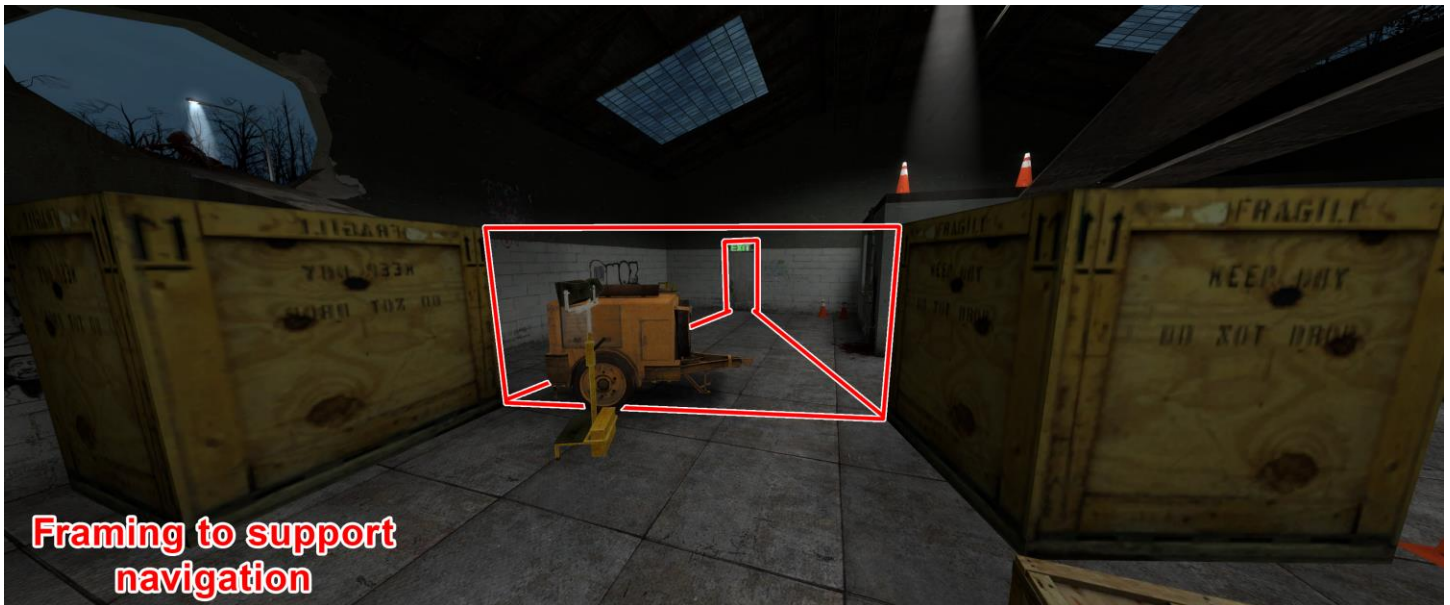
Colour Affordance

Colour affordance design is used to differentiate the main areas within the level. By assigning recognisable colours to portions of the level players can identify and memorise specific areas easier. The result is improved navigation and the ability to call out locations by an element that is both clear and specific to its location.



Framing

Framing has been used in specific applicable locations within the level to highlight potential pathways. As an approach improving the level's navigation, elements have been positioned to not block the view of pathways. Additionally, framing is used at the beginning of the level as a way of encouraging players towards the centre of the map upon spawning. This is with the goal of getting a majority of players to head this way as soon as the round starts as it's where the bulk of the map's elements are contained.



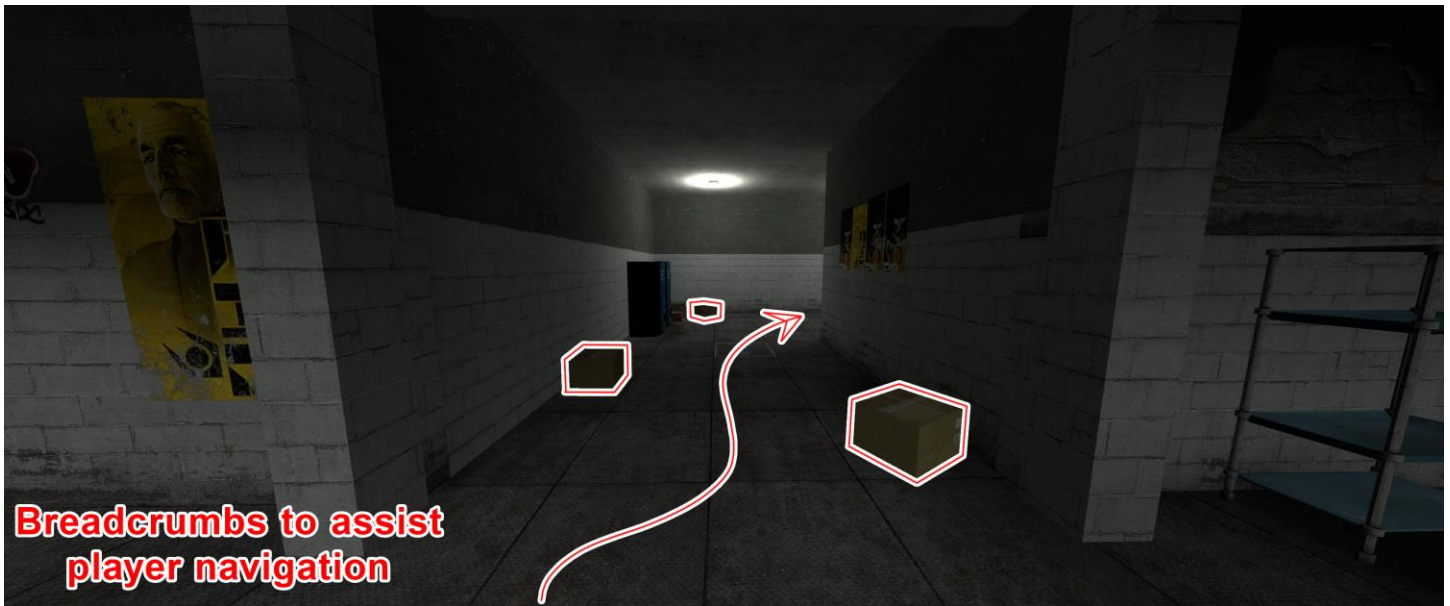
VFX

Visual effects such as blood, signs and other decals are used to create more affordant, consistent design within the level. Ladders and doors are an example of how VFXs are used to establish patterns within the level and a way of drawing attention to gameplay interactives. Elements nearby obvious VFXs tend to be interactive or interesting, this implementation remained throughout the level's usage of VFXs. Besides that, functionally similar to other affordances, where the intended implementation is to assist player behaviour by showcasing interactive level elements.



Breadcrumbs

Breadcrumbs are used throughout the level as a way to encourage players to take pathways. Props entities litter the pathways such as hallways. Players who will be checking each prop for Prop players will move down these pathways and into new area. Breadcrumbs are used in isolation at the beginning of the level to encourage players to move along towards the three major initial directions.



Semiotics

Semiotics within the level is used to visually describe the concepts of doors and ladders to players. Interactive elements need clear signifiers to detail players their simple usage, the visual models of doors and ladders achieve this. Doors have clear usage detailed through the handle, shape, and textures mimic real-world components of doors. Ladders follow an identical implementation with models placed in the location of ladder entities, that mimic ladders.



Sound

There is a singular usage of sound affordance used in the initial sections of the level. Near the red shed is a speaker entity connected to a TV prop which plays music. This implementation is with the goal of guiding players towards this location. Each of the three main directions has an implemented affordance to influence player behaviour towards the bulk of the level, this spot uses sound.



Anti Affordance

Anti-affordant techniques have been used to create clear boundaries for level. For the outside, fences surround the level, which are semiotic in nature, taller than players, and are an example of these clear boundaries. Additionally, thematic anti-affordances are used to create a blockade for the pathway leading out of the warehouse's carpark. This has the same effect as the fences, keeping player within, but is appropriate for the area of the level. Additional doors exist within the level to make the space seem larger, and to build on the narrative theme. These doors use anti-affordant techniques to create clear signifiers that these doors are not used within gameplay.



Level Narrative

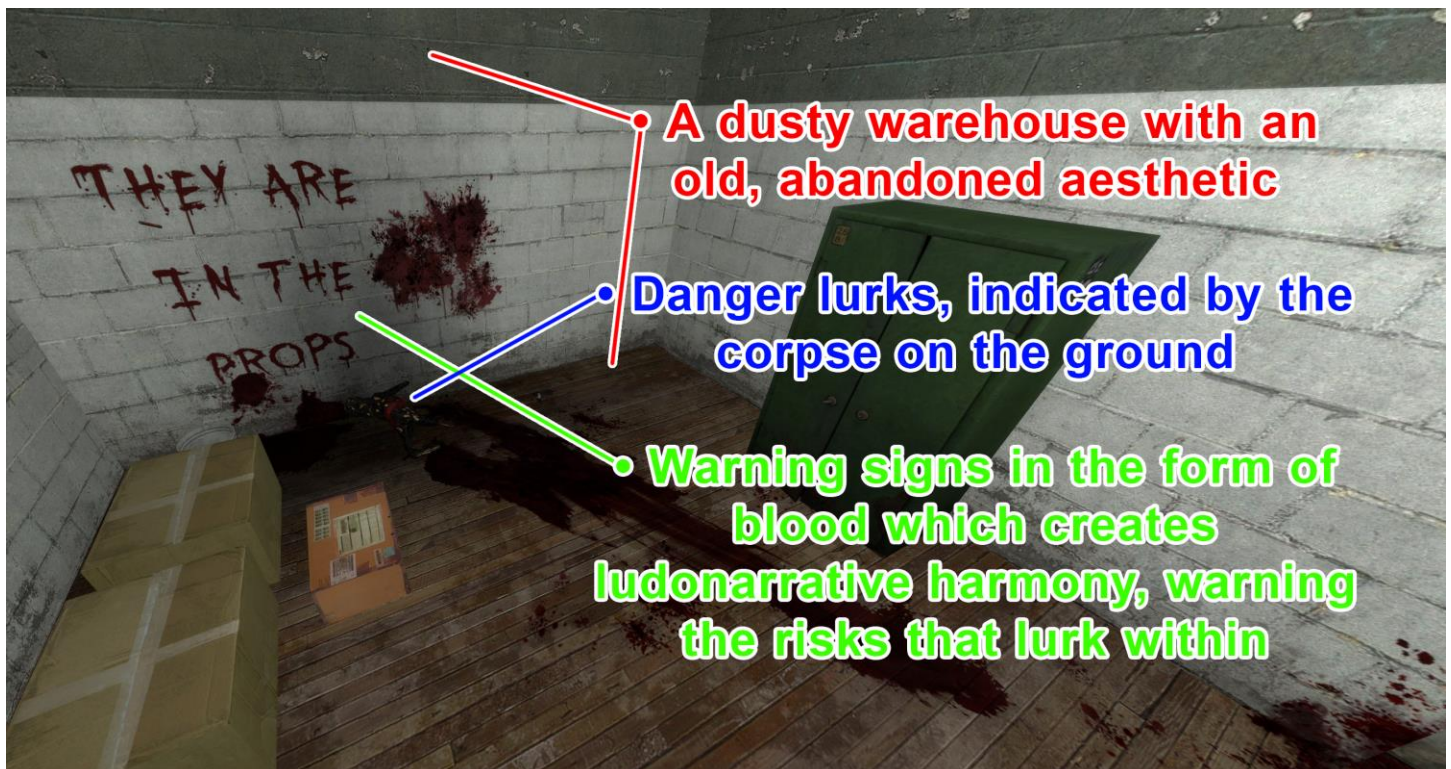
Level's Narrative Aesthetic

Level Narrative Overview

The level narrative is non-existent due to the gamemode's overall disconnect structure which focuses on environmental storytelling to reinforce the setting. Not all rules need to be followed, the importance of a rule decreases linearly in the list below.

- The backdrop of the level is an abandoned warehouse, which had a series of accidents occur within, causing the place to become lost to time. It's recently been rediscovered, with power being brought back to the place.
- But danger lurks within in the form of Prop Transformers who spread their malignant disease and torment those within to the brink of death. Corpses of those who had to experience their torture are spread throughout.
- Warning signs have been put up to warn people of the dangers within, but due to the risk that is all that has been placed. The lights still run in the place as people are too scared to get close to the warehouse again.

An example of the level narrative element must follow the example below as a basis. You have the basic aesthetic for the environment which is an abandoned warehouse; the room is old and dusty. Danger is lurking, with warning signs on the wall detailing the danger within. A clear sign that something is in here, and it's the props – this is a perfect way of creating ludo-narrative environmental set pieces.



Environmental Story Telling

Environment story telling set pieces have been used throughout the level to support the level's narrative and themes. The rules of environmental story telling have been sets, from which all narrative implementations follow.

For this example



For this example



Level Tone

Level Tone Overview

The tone of the level is multifaceted with a primarily tonally dark level, contrasted by a secondary, light-hearted tone with elements that players will find amusing throughout the level. This contrast should be used throughout the level in a ratio of 2:1. Two darker elements for one light-hearted one.

Optimal Level Testing

Testing for Prop Hunt

Testing Overview and Goals

Testing for the level should be focused on maintaining enjoyment above all else, gathering design feedback to improve on this aspect is the goal of testing the level. Participant's qualities are not important merits for how the design should be shaped, each volunteer is as valuable as the last. The testing was limited to six participants with teams of three, which will skew the data towards balancing and designing for that number of players. Each individual testing stage asks participants relevant questions about the level based on the current state.

- The first iteration was used to gather information about the design, balance, player enjoyment, memorability, and exploits. This was done with the express goal of gathering in-depth information about how players felt about the current state, what they enjoyed, disliked, which areas were interesting, and which were not.
- The second iteration's testing focused on similar aspects but asked less in detailed responses due to the changes not being massive from the previous version. Instead, this iteration focused on gathering a generalise set of feedback and allow participants to discuss areas they felt needed changes.